Lab-in-a-Droplet: Acoustic Levitation for Miniaturized Sample Preparation

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Motivation

- Sample Volume Constraints: Conventional clinical workup requires relatively large sample volumes, which are often not accessible
- Waste Accumulation: Routine sample throughput can accumulate high volumes of chemical and plastic waste
- Wall-Contact Losses: Conventional microvessels lead to sample loss through surface adsorption, and potential cross-contamination

Objective

- Need for Miniaturization: Adapting acoustic levitation as a green analytical platform, capable of handling contactless sample preparation
- Levitation Adaptation: Adapting conventional in-vial workup techniques within one levitating droplet (*in stillo*)
- Vitamin D Detection: Demonstrating feasibility, examplified by sample preparation of 25-hydroxyvitamin D in human serum

Methods

- Acoustic Levitation: Generation of an acoustic field through ultrasonic wave interference, leading to contactless suspension
- *in-stillo* Techniques: Protein precipitation, extractions and reactions within one levitating droplet through manipulation
- Analytical Detection: Manual collection and subsequent detection by liquid chromatography tandem mass spectrometry (LC-MS/MS)

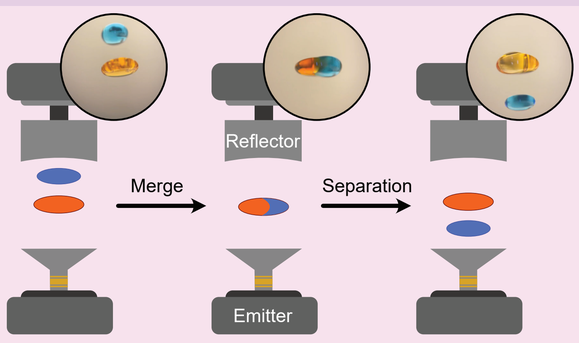
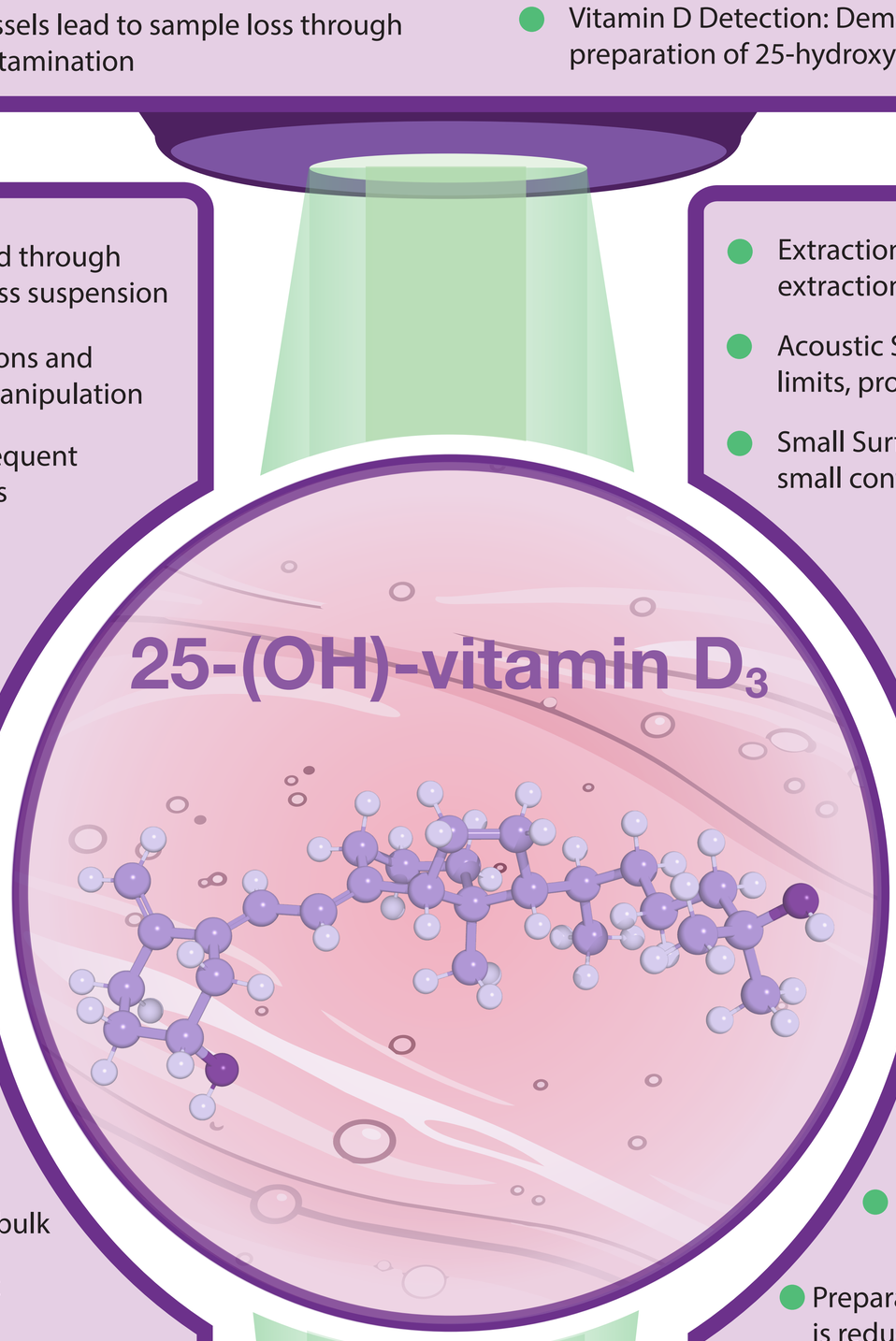


Figure 1: Schematic workflow representation.[1]



Sample Preparation

- Extraction Efficiency: Demonstrated contactless liquid–liquid extractions (*in stillo*) with high recovery rates
- Acoustic Streaming: Laminar internal flow overcomes diffusion limits, promoting rapid, contactless extraction (3 min)
- Small Surface Contact: Efficient extraction observed despite small contact interface between droplets

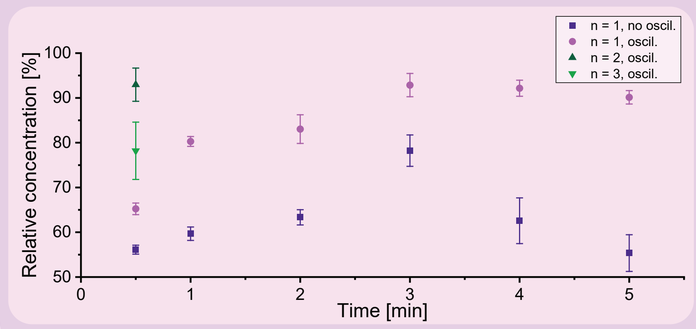


Figure 2: *in stillo* extraction of 25(OH)D_3.[1]

Fast Reaction

- Accelerated Kinetics: Acoustic streaming and high surface-to-volume promote reaction
- Surface Activity: Confined space promotes reaction at surface and back-diffusion into the bulk
- Evaporation: Preconcentration through solvent evaporation further accelerates reaction rate
- Product Formation: 60-fold increase in reaction rate observed, compared to conventional in-vial approach

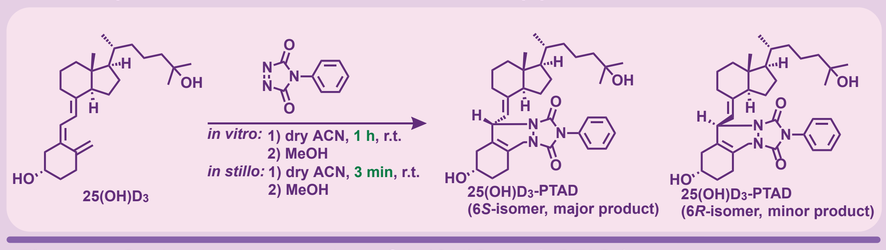


Figure 3: Derivatization reaction of 25(OH)D_3 with PTAD.[1]

Vitamin D Detection

- Comparable Metrics: Excellent linearity (R^2 = 0.997), LOQ = 2.9 ng/mL, 5.9% RSD, 15.4% agreement with *in vitro* assay
- Reduced Volumes: Approximately 90% reduction of required chemicals and sample
- Preparation Time: Overall sample preparation time is reduced from 90 min to 12 min

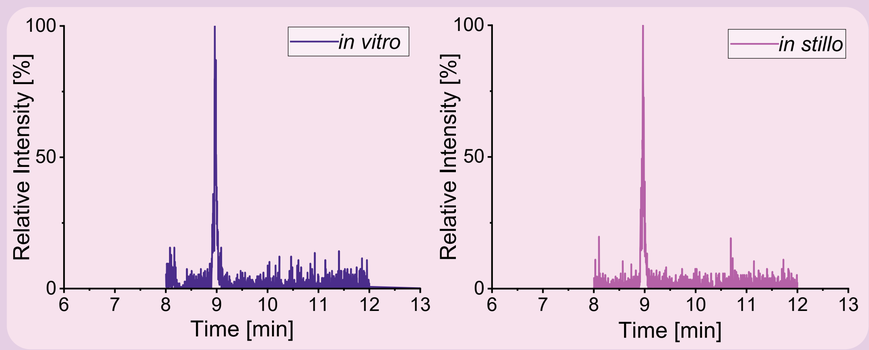


Figure 4: MRM chromatogram (m/z 558→298) for 25(OH)D_3 of human serum.[1]

Key Takeaway

- Proof-of-Concept Success: Demonstrated complete miniaturized sample preparation for 25-hydroxyvitamin D in human serum, yielding results comparable to conventional in-vial workflows
- Green Chemistry Platform: Offers a practical, low-cost green analytical chemistry framework for microliter analysis of biological samples
- Sample Handling: Manual sample handling proves challenging with current setup

Future Work

- Automation & High-Throughput: Automate droplet injection, mixing, and transfer, to improve reproducibility and high-throughput demand
- Expanded Applications: Adapting *in stillo* preparation workflows to other clinical biomarkers, toxins, and environmental pollutants
- Direct Instrument Interface: Integrate real-time ambient ionization mass spectrometry for instant, online reaction tracking and screening

References

Ratzek G.; Volmer D. A.; Lab-in-a-Droplet: Utilizing Acoustic Levitation for Miniaturized Sample Preparation and LC-MS/MS, Exemplified by Measurement of Vitamin D in Serum. *Anal. Chem.* **2026**; 98 (30): 22034–22042.

This work is funded by the Deutsche Forschungsgemeinschaft (DFG VO 1355/10–1).

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Levitate yourself!